

A Real-Time Multiplayer Social Touch System Using Computer Vision and Haptic Feedback

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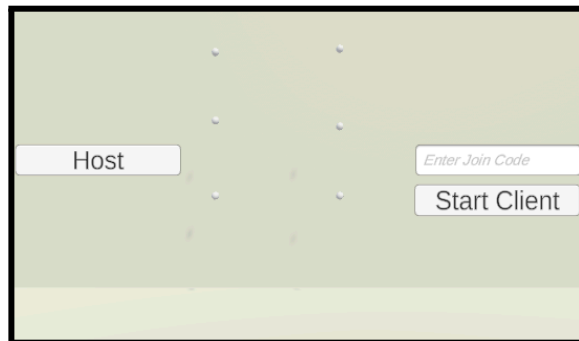
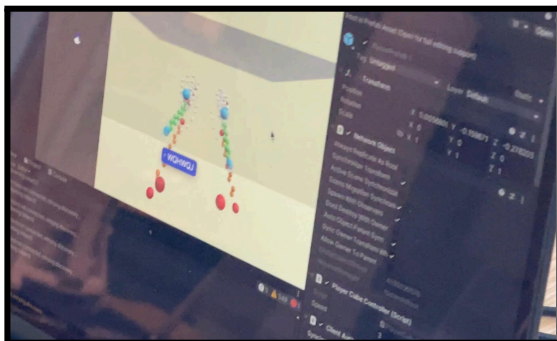
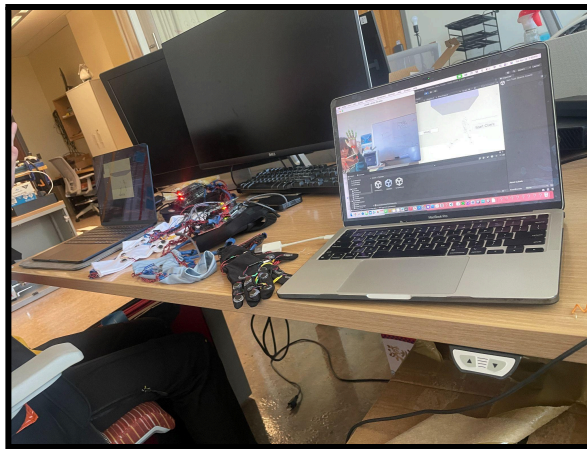
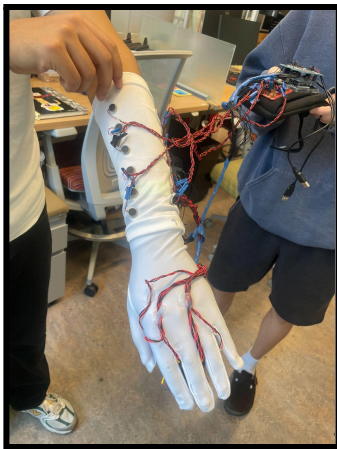
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Media:



(HaRVI Lab) Social Touch Project Software + Hardware Performance Test VIDEO:

<https://www.youtube.com/watch?v=smuGOeQZsMo>

BlazePose Tracking Demonstration:

<https://www.youtube.com/watch?v=GBjDifSEcul>

ABSTRACT

This project presents a real-time system for capturing human motion and translating it into shared virtual and physical interactions. Using computer vision-based pose estimation, multiplayer networking, and haptic hardware, the system enables users to experience “social touch” remotely. Arm and hand movements are tracked using BlazePose and hand landmark detection, mapped into a 3D environment, synchronized across multiple users, and converted into haptic feedback through embedded devices. This work explores how touch can be digitized, transmitted, and re-experienced across distance, contributing to research in embodied interaction and remote communication.

I. INTRODUCTION

Touch is a fundamental component of human communication, yet it is largely absent from digital interaction systems such as video calls and online environments. This project investigates how physical gestures can be captured, interpreted, and translated into meaningful touch interactions in a shared virtual space.

This work was developed in collaboration with the [HaRVI Lab](#) and focuses on enabling real-time, bidirectional interaction between users through motion tracking and haptic feedback.

Research Question:

Can computer-vision-based motion tracking combined with haptic feedback effectively simulate social touch between remote users in a shared virtual environment?

II. RELATED WORK

Prior research in human-computer interaction has explored gesture-based interaction and haptic feedback as mechanisms for enhancing remote communication. Vision-based tracking systems such as MediaPipe have enabled real-time pose estimation without specialized hardware, significantly lowering the barrier to motion capture.

In parallel, research in haptic systems has demonstrated the importance of tactile feedback in improving immersion and user experience in virtual environments. However, many existing systems rely on expensive motion capture setups or lack real-time bidirectional interaction.

This work builds upon these approaches by integrating real-time pose estimation, multiplayer synchronization, and haptic feedback into a unified system for social touch interaction.

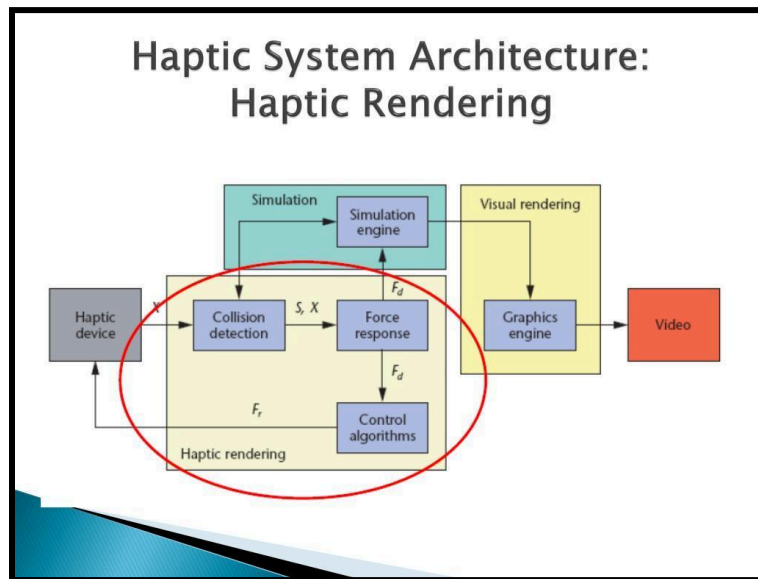
III. SYSTEM OVERVIEW

The system consists of four main components:

- Computer vision-based motion tracking
- 3D mapping and interaction in Unity
- Multiplayer synchronization
- Hardware-based haptic feedback

A Python pipeline captures motion data and transmits it via UDP to Unity. Unity reconstructs the motion in 3D space, synchronizes it across users, and triggers haptic responses based on interaction events.

Figure 1: System Architecture Pipeline



IV. POSE ESTIMATION AND GESTURE TRACKING

The system uses MediaPipe's **BlazePose model** for real-time pose estimation.

BlazePose provides normalized 2D joint coordinates and relative depth estimates for key body landmarks. For this prototype, the system focuses on upper-limb tracking by extracting:

- Shoulder joints (indices 11, 12)
- Elbow joints (indices 13, 14)
- Wrist joints (indices 15, 16)

These joints are selected due to their importance in representing intentional touch gestures such as reaching, tapping, and pressing.

The pose estimation pipeline runs in Python using OpenCV for frame acquisition and preprocessing.

V. 3D MAPPING AND DEPTH APPROXIMATION

Joint positions are mapped from normalized image space into Unity world coordinates using a camera-relative coordinate system.

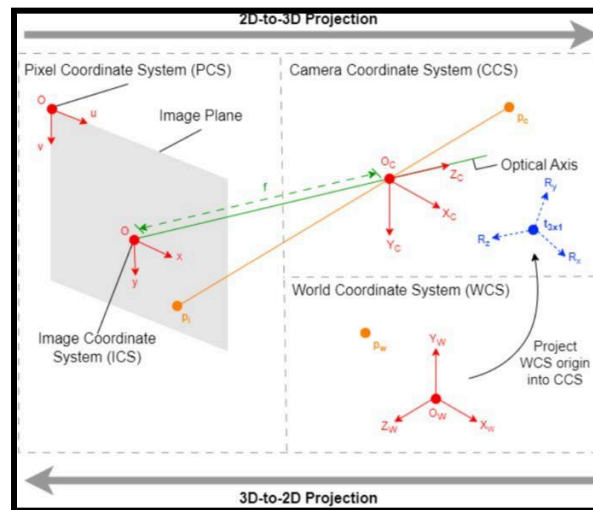
Depth is approximated using two components:

- **Relative depth ($z_{\square}$)** from BlazePose
- **Distance proxy** derived from shoulder width

As the user moves farther from the camera, the apparent shoulder width decreases. As this happens, user distance increases, which enables depth inference. This inverse relationship is used to estimate depth and adjust the reconstructed position in 3D space.

This method produces a stable pseudo-3D representation without requiring specialized depth sensors.

Figure 2: Coordinate Mapping



VI. INTERACTION AND GESTURE REPRESENTATION

Tracked joints are used to infer interaction primitives:

- **Collision-based contact** → touch events
- **Penetration depth** → pressure intensity
- **Motion trajectories** → gesture dynamics

Users are represented as joint-based spheres within Unity, enabling collision detection between participants and virtual objects.

VII. MULTIPLAYER NETWORKING

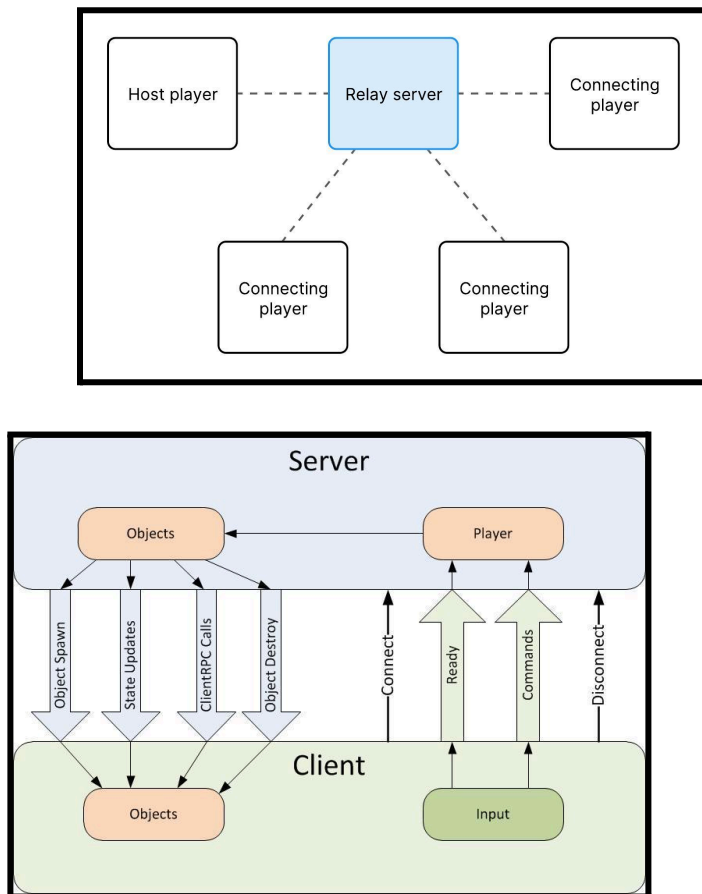
The system supports real-time interaction between users using Unity Netcode for GameObjects.

Each client:

- Processes motion data locally
- Synchronizes joint positions across the network
- Receives and renders remote user motion in real time

This architecture ensures low latency and enables shared spatial interaction between participants.

Figure 3 & 4: Networking & Synchronization



VIII. HAPTIC FEEDBACK INTEGRATION

Touch interactions are extended into the physical world through custom hardware.

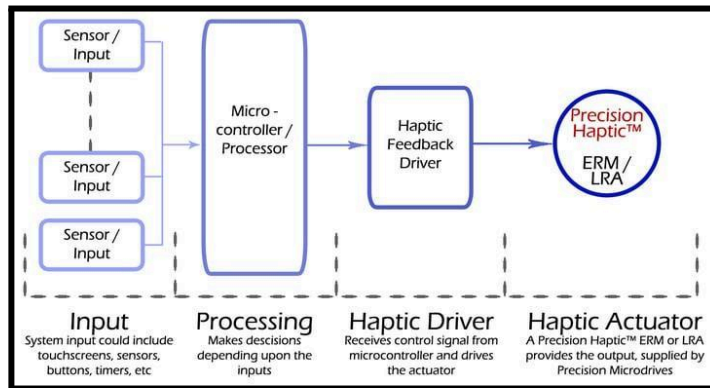
- Collision events generate haptic signals
- Signals are transmitted via TCP to microcontroller-based devices
- Actuators provide feedback based on:

- Contact intensity
- Duration
- Interaction location

This creates a bidirectional interaction loop:

physical movement → digital interaction → physical sensation

Figure 5: Haptic Loop



IX. Evaluation

To evaluate the system, performance and interaction quality were analyzed across multiple sessions.

Latency: Real-time interaction was maintained with low perceptible delay between motion and response

Stability: Joint tracking remained consistent under typical lighting conditions, with minor jitter during occlusion

Interaction Quality: Users were able to reliably trigger touch interactions through collision-based gestures

Preliminary observations indicate that combining pose estimation with haptic feedback enhances the perception of interaction compared to visual-only systems.

X. Limitations

Several limitations were identified:

- Depth estimation relies on heuristics rather than true depth sensing
- Occlusion can reduce tracking accuracy
- Network latency may affect synchronization between users
- MediaPipe output introduces minor positional noise

These factors may impact the realism of the interaction.

XI. Applications

The system demonstrates potential across multiple domains:

Enhanced Video Communication:

Integration of tactile feedback into video calls enables more expressive and immersive remote interaction.

Shared AR/VR Environments:

Real-time motion tracking and haptic feedback support collaborative virtual spaces where users can physically interact with shared objects.

Remote Collaboration Systems:

The system can be used in teleoperation, remote training, and cooperative tasks where physical interaction improves communication.

Social and Emotional Communication:

Haptic feedback enables the transmission of gestures such as taps or touches, supporting more natural and emotionally expressive interaction.

Motion Capture for Interactive Media:

The pipeline provides a lightweight alternative to traditional motion capture systems, enabling real-time body tracking for games, animation, and virtual production without specialized equipment.

Gaming and Interactive Entertainment:

The system can enhance gameplay by incorporating full-body interaction and physical feedback, enabling new mechanics based on touch, gesture, and proximity.

XII. Challenges

Several challenges were encountered during the development of the system, particularly in integrating hardware with vision-based tracking.

One major challenge involved the **visibility of the haptic glove**. The initial goal was to train a custom machine learning model capable of recognizing the glove and its embedded actuators. However, this approach proved to be time-intensive and difficult to generalize due to variations in background environments and lighting conditions, which significantly impacted model performance.

Additionally, existing hand-tracking solutions such as MediaPipe are optimized for detecting natural skin features and struggle to reliably track hands when covered by hardware components.

To address this limitation, the system transitioned to using **BlazePose-based upper-body tracking**, which focuses on larger skeletal joints (shoulder, elbow, wrist) and is more robust to occlusion and visual variation.

As a hardware design solution, a glove covering was introduced to conceal actuators and create a more uniform visual appearance, improving tracking consistency while maintaining functionality.

Despite these solutions, challenges related to occlusion, lighting variability, and the lack of precise depth sensing remain limitations of the current system.

XIII. Conclusion

This project demonstrates a complete pipeline for capturing, transmitting, and reproducing touch interactions across distance. By combining computer vision, networking, and hardware systems, it enables a new form of communication centered around embodied interaction.

Future work may include improved depth estimation, machine learning-based gesture recognition, and more advanced haptic devices.
